MIL-C-45408B(MU)
5 April 1972
SUPERSEDING
MIL-C-45408A(MU)
19 August 1963

MILITARY SPECIFICATION

CARTRIDGE, IMPULSE: M44A1 ASSEMBLY

1. SCOPE

1.1 This specification covers Cartridge, Impulse: M44A1 for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-C-60539	- Coating, Anodic, Conventional, for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MS28741	- Hose Assembly, Detachable End Fitting, Medium Pressure

FSC 1377

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2.1 (Cont'd)

DRAWINGS

Munitions Command

D8593276	- Cartridge, Impulse, M44A1, Assembly
C8797441	- Box, Packing, Ammunition for Cartridge, Impulse M44A1
C8797373	- Carton, Packing, Ammunition for Cartridge, Impulse, M44A1
IEL-8593276	- Inspection Equipment List for Cartridge, Impulse, M44A1 Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI) 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Vibration.-The assembled cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant or derangement of any component.

3.3 Protective finish.-Protective finish shall be in accordance with MIL-C-60539.

3.4 Functional.

3.4.1 The cartridge shall not fail to fire.

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3.4.2 Ballistics.-The cartridge, when fired in an M3A3 Thruster, temperature conditioned at -65°F to 160°F shall produce an effective thrust to assure a minimum piston stroke of 1.5 inches when propelling a 550 pound mass vertically upward at a velocity such that the propelled load continues to move upward after the piston reaches the end of its stroke. It shall develop a by-pass pressure of 600 pounds per square inch (psi), minimum, at the end of 4 feet of Aircraft hose, Size No. 4 MS28741.

3.4.2.1 Initiation.-The thruster shall be initiated by a source delivering an output equal to an M3A1 Initiator at the end of 15 feet of aircraft hose, Size No. 4, MS28741 at 70° ± 5°F.

3.5 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.-The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.5.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, rust, oil, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

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4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 21 assembled cartridges for ballistic test, 5 assembled cartridges for sealant application test and 10 sets of unassembled components for visual and dimensional examination and tests. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used for production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.-After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.-The sample for ballistic tests shall be temperature conditioned and conform to requirements as indicated in 3.4.2. The sample for tests shall consist of the following:

TABLE I

Temperature	Not Vibrated	Vibrated
-65°F to -70°F	3	4
65°F to 75°F	3	4
160°F to 165°F	3	4

4.2.1.3 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.-An ammunition lot shall consist of metal parts assemblies consisting of:

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4.3.1.2 (Cont'd)

a. Component produced by a homogeneous process.
b. Propellant from not more than one lot.
c. Igniter from not more than one lot.
d. Primers from not more than one lot.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables II and III. For an individual defect (single characteristic) the criteria in Table II shall be used. For total defects within a major or minor group the criteria in Table III shall be used. All non-conforming material shall be rejected. The roughness comparison specimens prescribed by ANSI B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE III

Total Allowable Defects per Sample Size

Sample Sizes for Major Defects	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14

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4.3.2 (Cont'd)

TABLE III (Cont'd)

Sample Sizes for Minor Defects	110	150	225	300
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.3.2.1 Primer, Percussion, Drawing D10551688 covering a detail of Dwg. D8593276.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
2. Primer foil discolored	Visual
3. Mixed primers	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.3.2.2 Cartridge Head, Drawing D8594159 covering a detail of Drawing D8593276.

CRITICAL	
1. Depth to base of small counterbore	Gage
2. Depth of primer pocket	Gage
3. Metal defective	Visual

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Concentricity of small counterbore with primer pocket	Gage
102. Concentricity of flash hole with primer pocket	Gage
103. Concentricity of flange head with body	Gage
104. Concentricity of flange head with primer pocket	Gage
105. Diameter of cartridge head recess, min.	Gage
106. Diameter of body, min.	Gage
107. Protective finish inadequate or defective	Visual
108. Surface finish of primer pocket	Visual
109. Burr, except where permitted	Visual
110. Foreign matter	Visual
MINOR	
201. Diameter of counterbore at head	Gage
202. Diameter of body, max.	Gage
203. Diameter of flash hole	Gage
204. Diameter of flange head	Gage
205. Diameter of cartridge head recess, max.	Gage
206. Depth of counterbore at head	Gage
207. Total length	Gage
208. Length of chamfer at flash hole	Gage
209. Location of datum diameter	Gage
210. Location of large diameter of flange	Gage
211. Angle on head	Gage
212. Surface finish improper, except as otherwise classified	Visual

4.3.2.3 Case, Cartridge, Drawing C8594158 covering a detail of Drawing D8593276.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Straightness of base	Gage
103. Diameter of mouth	Gage

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4.3.2.3 (Cont'd)

Defect	Method of Inspection
MAJOR	
104. Thickness of base	Gage
105. Thickness of sidewall	Gage
106. Protective finish inadequate or defective	Visual
107. Burr or sharp corner	Visual
108. Foreign matter	Visual
MINOR	
201. Diameter of body	Gage
202. Length of sidewall at mouth	Gage
203. Total length	Gage
204. Location of datum diameter	Gage
205. Radius or chamfer missing or incorrect	Visual
206. Surface finish improper	Visual

4.3.2.4 Assembly, Cartridge Impulse M44A1 Assembly Drawing D8593276.

CRITICAL	
1. Depth to primer pocket face	Gage
2. Primer not below flush with surface "S"	Gage
3. Primer pellet broken or dislodged	X-Ray & Visual 1/
4. Any part missing	X-Ray & Visual 1/
5. Improper assembly	X-Ray & Visual 1/
6. Primer not fully seated in primer pocket	X-Ray & Visual 1/
7. Crimp missing or incorrect	Visual
8. Mixed ammunition	Visual
9. Marking missing, incorrect or illegible	Visual
10. Metal defective	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Length to datum diameter, max.	Gage
103. Burr	Visual
104. Foreign matter	Visual

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4.3.2.4 (Cont'd)

Defect	Method of Inspection 1/
MINOR	
201. Total length, max.	Gage
202. Draw scratch	Visual
203. Fold, wrinkle, buckle, bulge or dent	Visual
204. Varnish missing or inadequate	Visual

1/ The cartridge shall be X-rayed with the sides of the cartridge perpendicular to the rays. The X-ray negatives properly identified shall be forwarded to the procuring agency.

4.3.2.5 Carton, Packing, Ammunition for Cartridge, Impulse, M44A1, Drawing C8797373.

CRITICAL

None defined

MAJOR

None defined

MINOR	
201. Contents loose	Manual & Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Heat sealer incomplete, loose or distorted	Visual
205. Cut or damage through impregnated	Visual
206. Marking missing, incorrect or illegible	Visual

4.3.2.6 Box, Packing, Ammunition for Cartridge Impulse, M44A1, Drawing C8797441.

CRITICAL

None defined

MAJOR

None defined

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4.3.2.6 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual & Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Vibration.-Twelve of the 21 cartridges of the initial production sample shall be subjected to vibration. The 12 cartridges shall then be x-rayed. Failure of any cartridge to comply with the requirements of 3.2 shall be cause for rejection of the lot. Any change in the process of a lot of primers in subsequent production shall be reason to select an additional sample of 12 cartridges for vibration tests.

4.3.3.2 Protective finish.-Unless otherwise specified, sampling plans and procedures in determination of the acceptability of coated parts and articles submitted by a supplier shall be in accordance with provisions set forth in MIL-STD-105.

4.3.3.3 Airtightness.-Each cartridge shall be subjected to the specific pressure differential. Failure to comply with the requirements of the drawing shall be cause for rejection of the cartridge.

4.3.3.4 Sealant application.-Five cartridges from each production lot shall be selected for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot of cartridges.

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4.3.3.5 Hardness.-Fifty units from each heat treated lot of components shall be tested. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.6 Functional.-The following test shall be performed by the Government:

4.3.3.6.1 Ballistics.-Twenty-one cartridges from each production lot shall be grouped and temperature conditioned as prescribed in Table IV. Failure of any cartridge to comply with 3.4 shall be cause for rejection of the lot of cartridges.

TABLE IV

Temperature	Sample Size
-65°F to - 70°F	7
65°F to 75°F	7
160°F to 165°F	7

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL-8593276.

4.4 Test methods and procedures.

4.4.1 Vibration.-The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.2 Protective finish.-Test procedures for protective finish shall be in accordance with MIL-C-60539. The cartridge case and head shall be sectioned longitudinally to better expose the significant surfaces to the salt spray.

4.4.3 Airtightness.-The method for determining airtightness shall be by use of equipment listed on IEL-8593276.

4.4.4 Sealant application.-The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length of the joining surfaces at the junction of the cartridge head and case sidewall and that the propellant cavity is free of sealant.

4.4.5 Hardness.-A hardness testing machine shall be used to determine hardness requirements specified on drawings.

4.4.6 Functional.

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4.4.6.1 Ballistics.-The cartridges shall be loaded into an acceptable M3A3 Thrusters, divided into 3 equal groups and temperature conditioned in accordance with Table IV. The cartridge shall be tested within 5 minutes after removal from the conditioning chamber. When this time is exceeded, the catapult shall be reconditioned in accordance with 4.4.6.1.1.

4.4.6.1.1 Temperature conditioning.-A simulated M3A3 Thruster loaded with a inert M44A1 cartridge, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with Drawings C8797373 and C8797441.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Hazard notice.-The cartridge described herein and certain of its components are flammable and/or explosive, consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

6.3 Loaded Assembly Hazard Information:
Explosive Hazard Classification: Class 3
Storage Compatibility: Group B
Department of Transportation (DOT) Hazard Class C
DOT Marking - Explosive Power Devices - Class C
Handle Carefully - (Keep Fire Away)

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1377-A503

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SPECIFICATION ANALYSIS SHEET

Form Approved.
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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